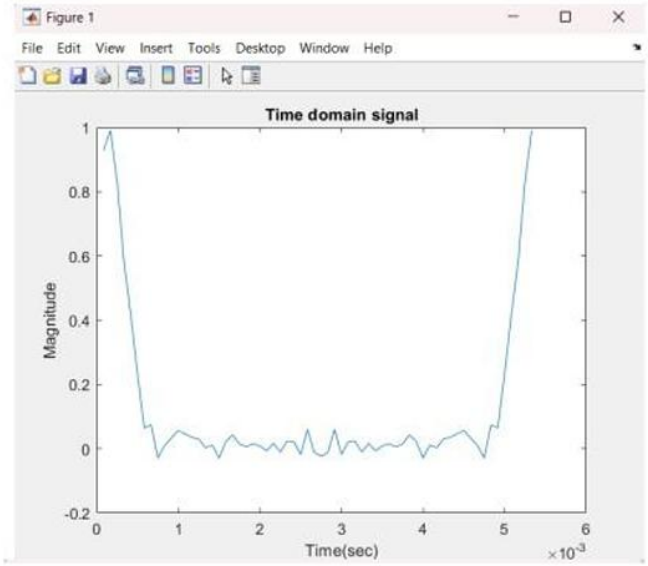
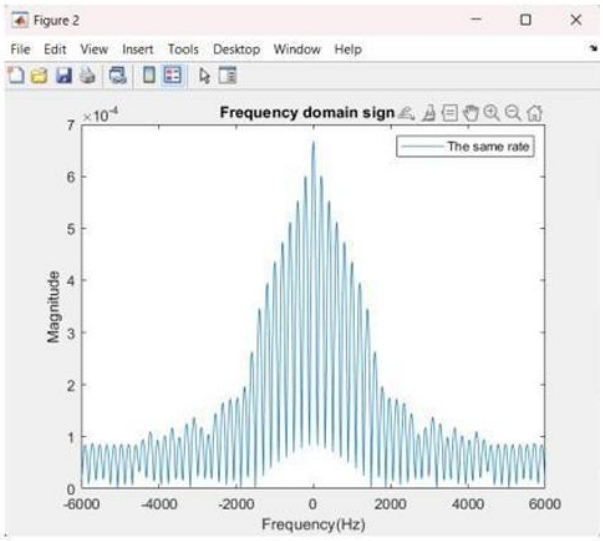
DSP Lab Report E4

Names	IDs	Section	Group No. UG2023
Amr Hatem Sabry Elmancy	2300729	1	29
Abdelrahman sherif	2300100	1	
Abdelrahman Mohamed Ali	2300518	1	

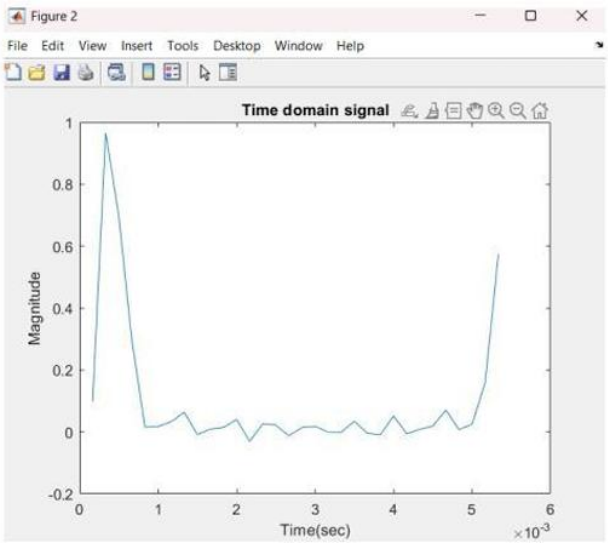
1) **Apply decimation to the vector by different decimation rates (2,4) with anti_aliasing filter. Plot the wave form of the original signal and the decimated signal for both the rates. Plot the spectrum of the decimated signal for different rates and compare those spectrums to the original spectrum**

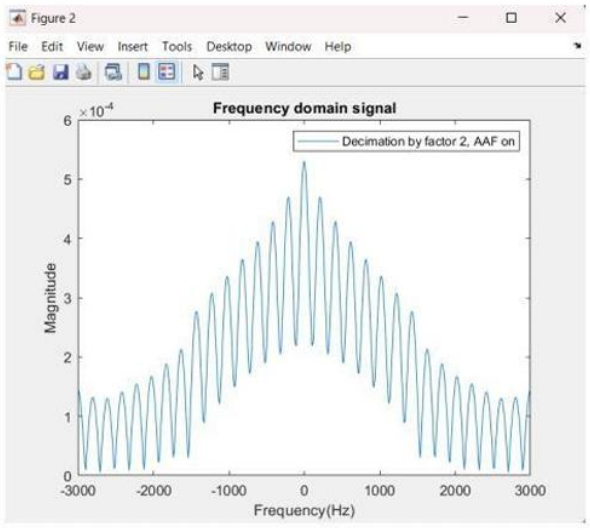
original signal:



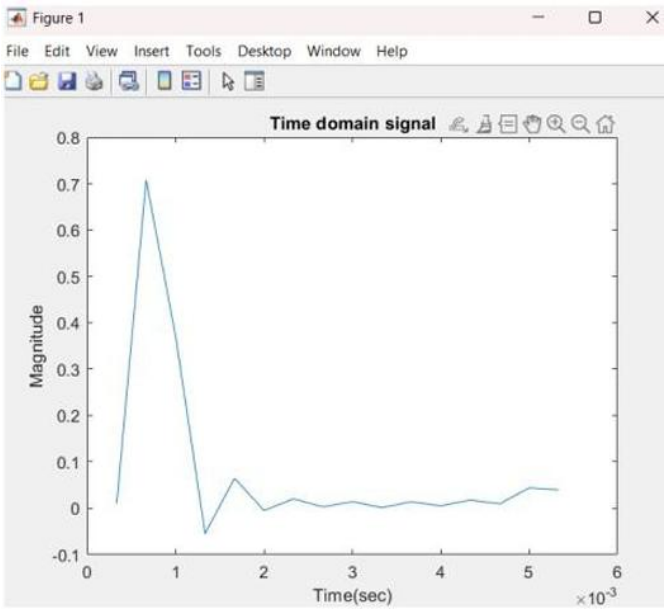


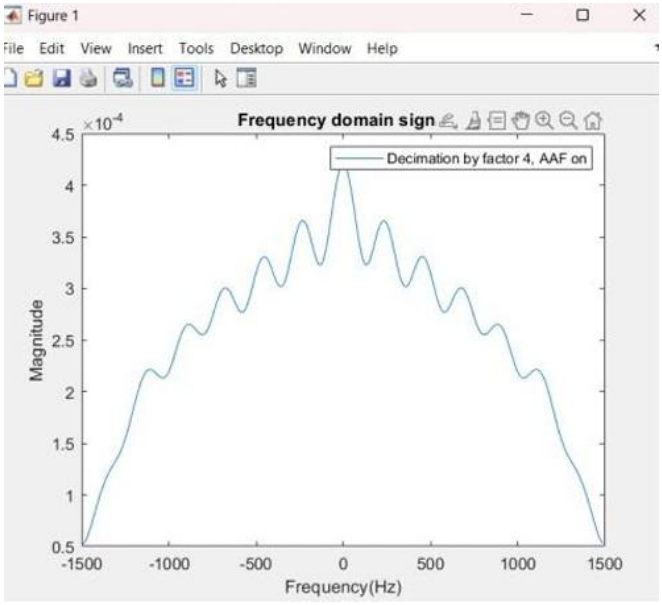
Applying decimation by (2) with anti-aliasing filter:





Applying decimation by (4) with anti-aliasing filter:





Decimation Rate = 2

Time Domain

- The new signal looks very similar to the original one.
- The amplitude and general shape stay the same.
- Even though there are fewer data points, the lines still look smooth.

Frequency Domain

- The frequency matches the original signal for the first half.
- High frequencies have been cleaned out.
- Because of the filter, there is no aliasing in the data.

Decimation Rate = 4

Time Domain

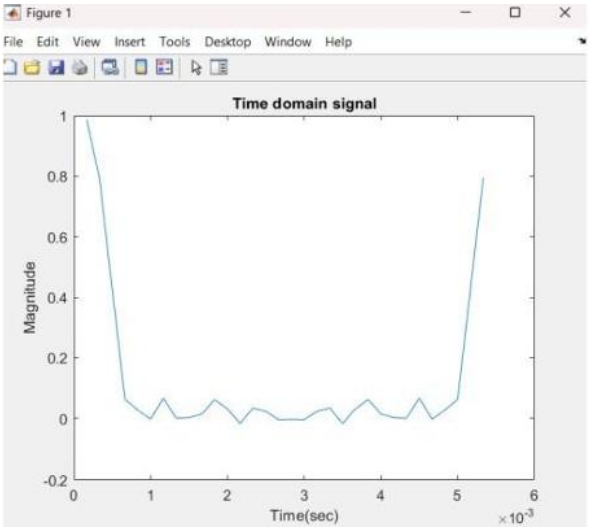
- The signal still has the same basic shape, but it looks "tighter" or more squashed.
- Some of the tiny details are gone because we are taking fewer measurements.
- You can still tell what the signal is, but it has lower detail.

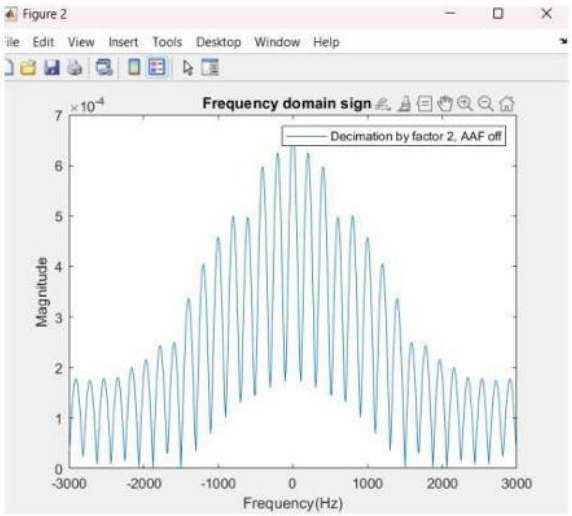
Frequency Domain

- Only the very lowest frequencies are kept.
- The graph looks much thinner than the Rate 2 version.
- The filter worked well, so there is still no distortion from aliasing.

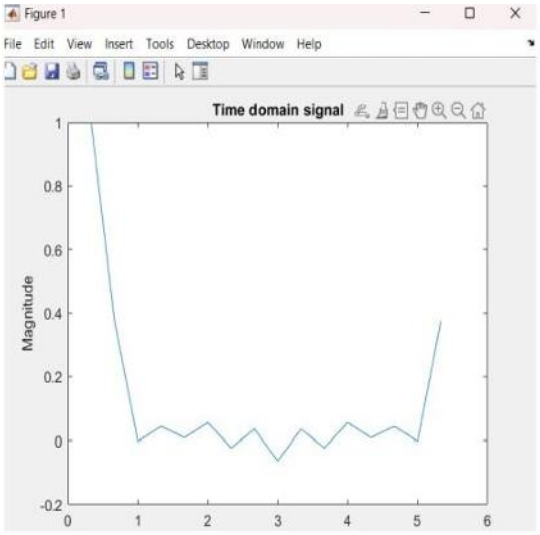
2) Apply decimation to the vector by different decimation rates (2,4) without anti_aliasing filter. Plot the wave form of the original signal and the decimated signal for both the rates. Plot the spectrum of the decimated signal for different rates and compare those spectrums to the original spectrum

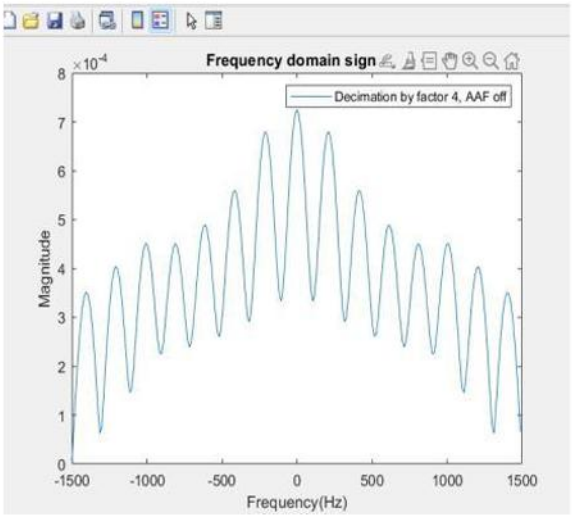
Applying decimation by (2) without anti-aliasing filter:





Applying decimation by (4) without anti-aliasing filter:





Decimation Rate = 2

Time Domain

- The signal looks rough and is not as smooth as the original.
- The edges and transitions look jagged or inconsistent.

Frequency Domain

- You can start to see "aliasing" (distortion) happening.
- High-frequency energy is "folding over" into the lower frequencies.
- You see extra, unwanted peaks that shouldn't be there, showing the signal is getting messy.

Decimation Rate = 4

Time Domain

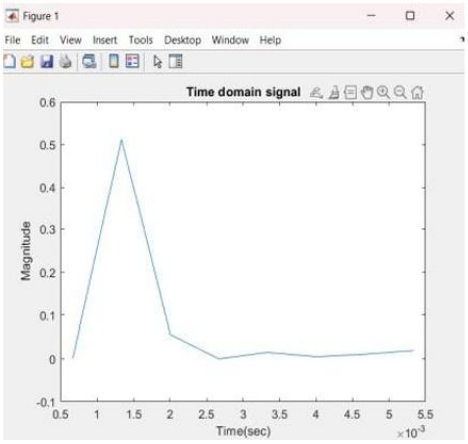
- The signal is heavily distorted and looks broken.
- It is hard to recognize the original shape or "envelope" of the signal anymore.

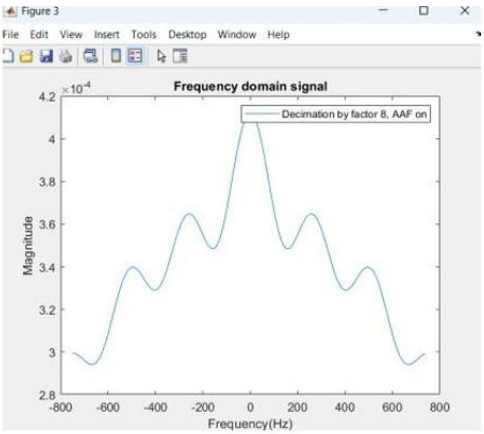
Frequency Domain

- Aliasing is very obvious now, with many strong, unwanted parts appearing in the frequency.
- Different parts of the spectrum are overlapping, which ruins the signal.
- The frequency data is completely wrong and no longer matches the original.

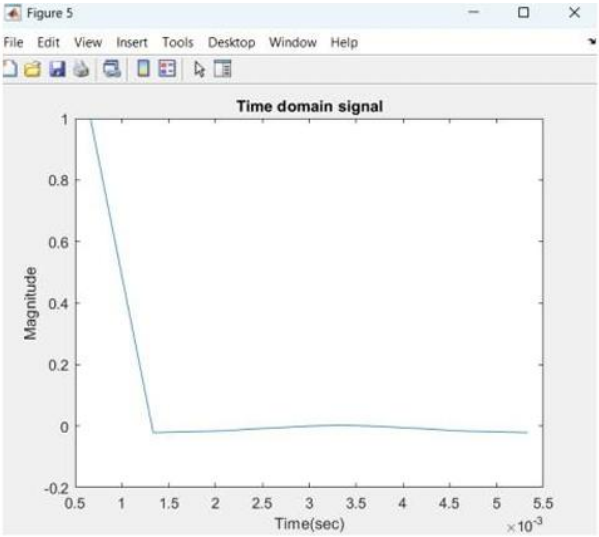
3) Apply decimation to the vector by decimation rate equals (8) with and without anti_aliasing filter. Plot the wave form of the original signal and the decimated signal for both cases. Plot the spectrum of the decimated signal for different rates and compare those spectrums to the original spectrum

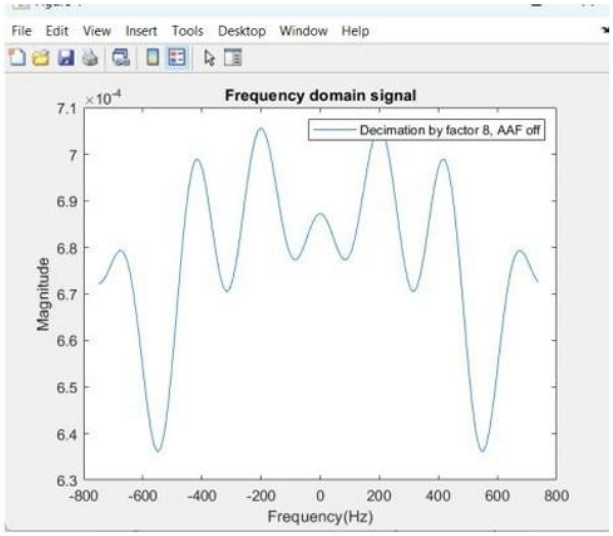
Applying decimation by (8) with anti-aliasing filter:





Applying decimation by (8) without anti-aliasing filter:





Decimation Rate = 8 (With Filter)

Time Domain

- The signal looks clean and smooth, but it has much less detail because there are very few data points left.
- You can still see the basic "skeleton" or main shape of the original signal.

Frequency Domain

- The low-frequency parts (the main parts of the signal) are still strong.
- Almost all high-frequency parts have been removed to keep the signal from getting messy.

Decimation Rate = 8 (Without Filter)

Time Domain

The signal looks random, jumpy, and jagged. It doesn't make much sense visually.

- You can no longer tell what the original shape was supposed to be.

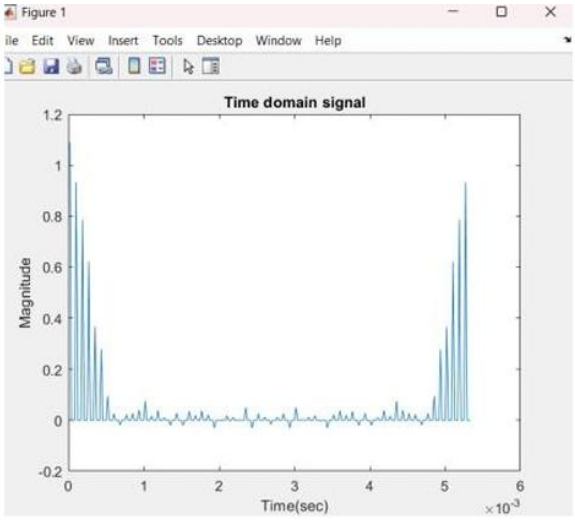
Frequency Domain

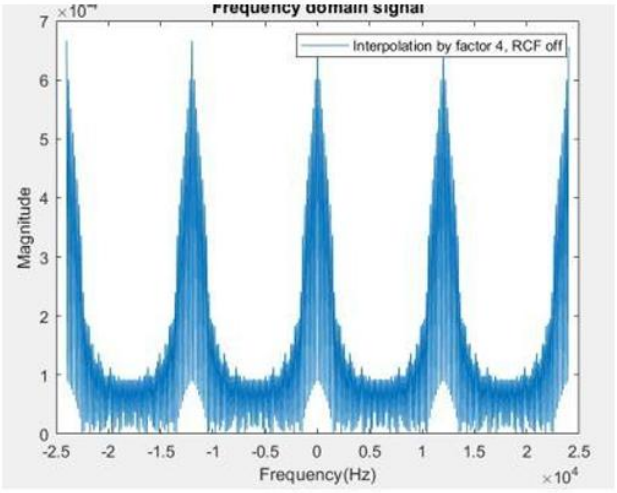
The spectrum is completely ruined by severe aliasing.

- Different frequencies are crashing into each other, creating "ghost" versions of the signal (mirrored artifacts).
- Fake peaks appear everywhere, making it look like the signal contains information that was never actually there.

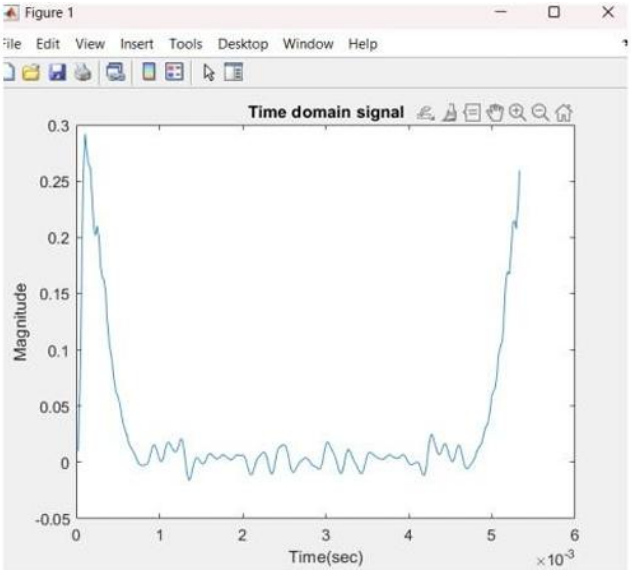
4) **Apply interpolation to the vector by different interpolation rates (4,8) with and without reconstruction filter. Plot the waveform of the original signal and the interpolated signal for both the rates. Plot the spectrum of the interpolated signal for different rates and compare those spectrums to the original spectrum.**

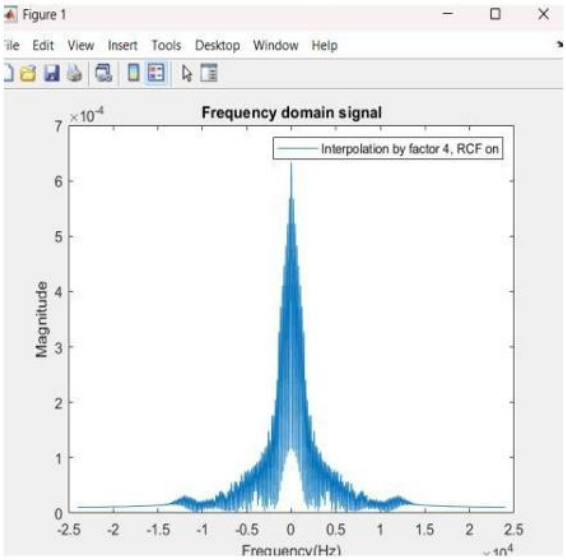
Applying interpolation by (4) without reconstruction filter:



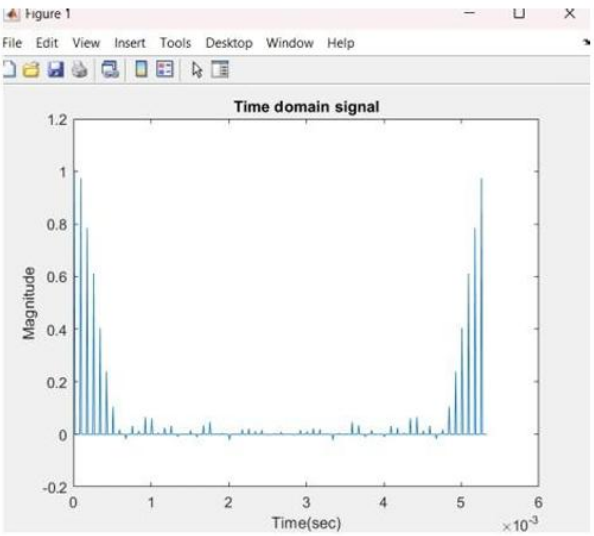


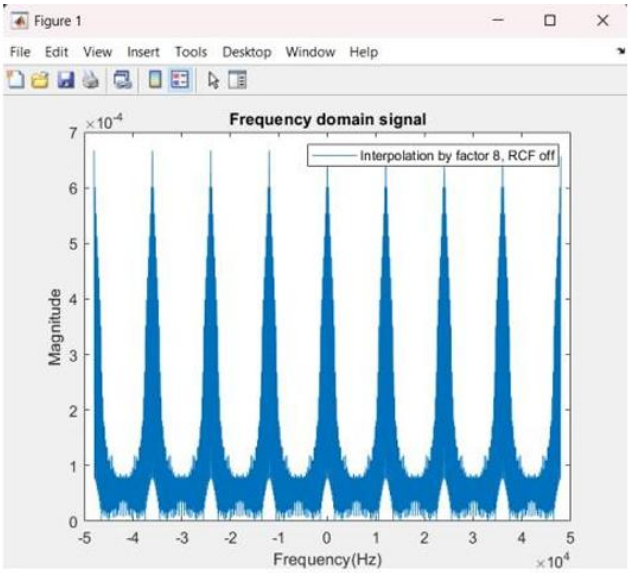
Applying interpolation by (4) with reconstruction filter:



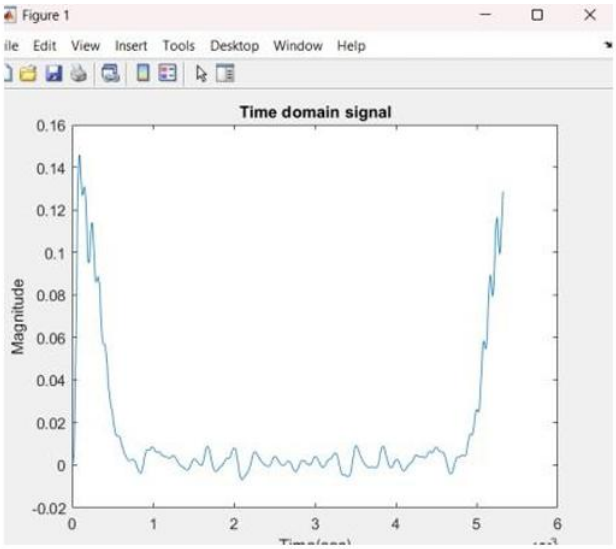


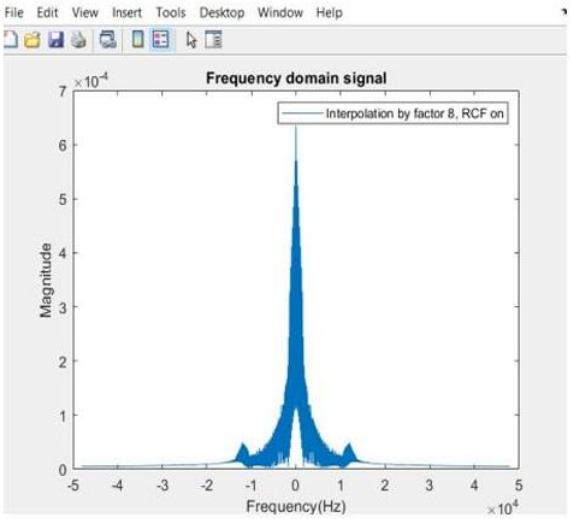
Applying interpolation by (8) without reconstruction filter:





Applying interpolation by (8) with reconstruction filter:





Interpolation Rate = 4

Time Domain

- **With Filter:** The signal looks very smooth and connected. It is a very close match to the original shape.
- **Without Filter:** The signal looks like a staircase. This is because we added extra space (zeros) between the original points, creating sharp, blocky steps.

Frequency Domain

- **With Filter:** The spectrum looks natural and clean. There are no weird, extra signals added.
- **Without Filter:** You see "spectral replicas." These are extra copies of the signal's original frequency appearing where they don't belong.

Interpolation Rate = 8

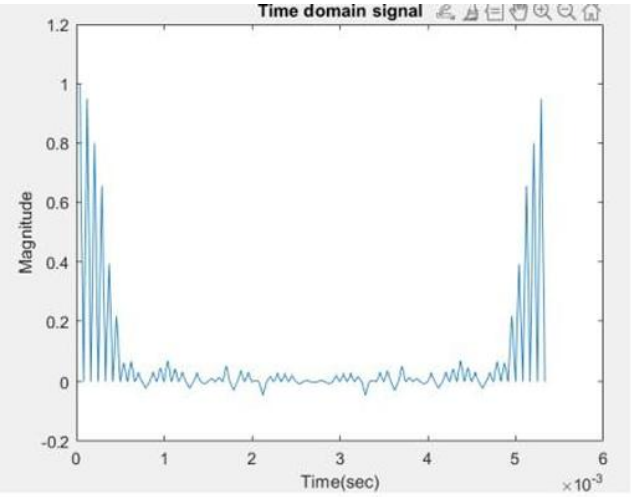
Time Domain

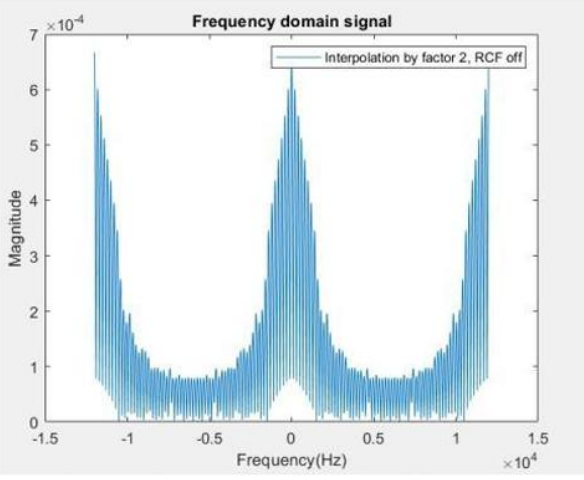
- **With Filter:** The signal is extremely smooth and full of detail. It looks almost identical to a high-quality version of the original.
- **Without Filter:** There are long gaps of "nothing" (zeros) between the actual data points. This makes the signal look empty and broken.

Frequency Domain

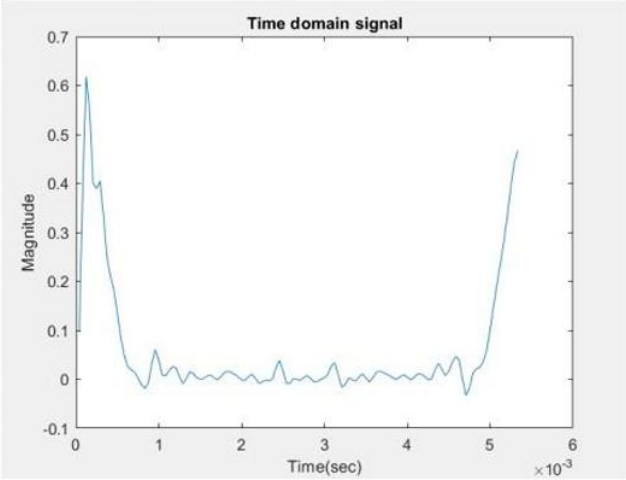
- **With Filter:** The spectrum stays clean and stays within its original limits.
- **Without Filter:** Many crowded "images" or copies of the signal appear in the frequency graph. These extra parts ruin the accuracy of the signal.

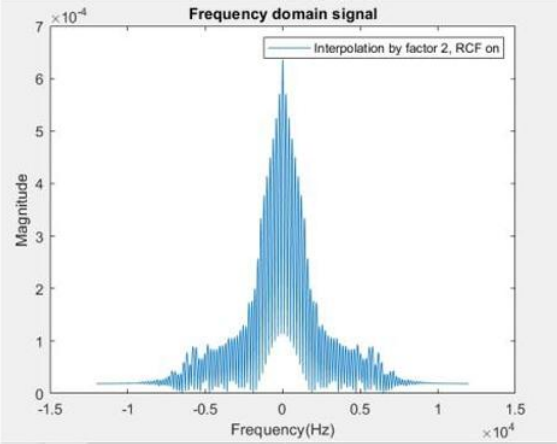
5) **Apply interpolation to the vector by different interpolation rates (2,16) with and without reconstruction filter. Plot the waveform of the original signal and the interpolated signal for both the rates. Plot the spectrum of the interpolated signal for different rates and compare those spectrums to the original spectrum**
Applying interpolation by (2) without reconstruction filter:



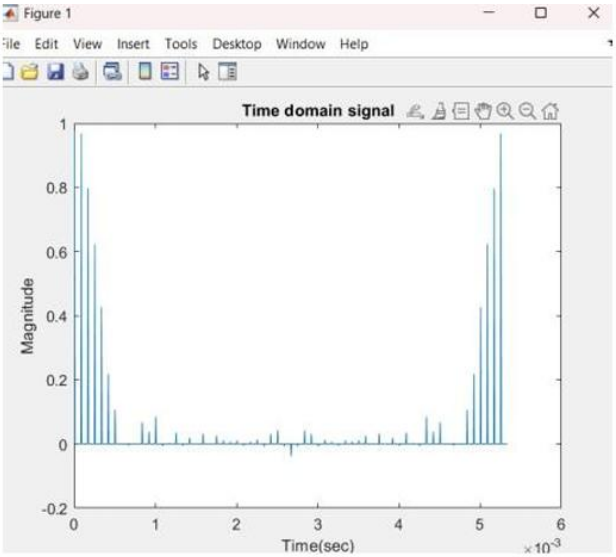


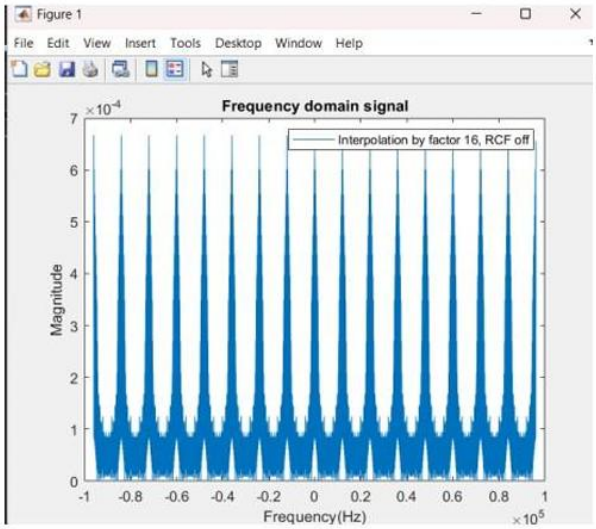
Applying interpolation by (2) with reconstruction filter:



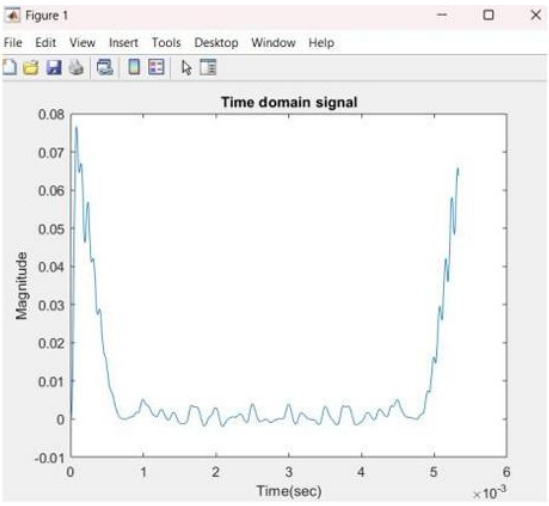


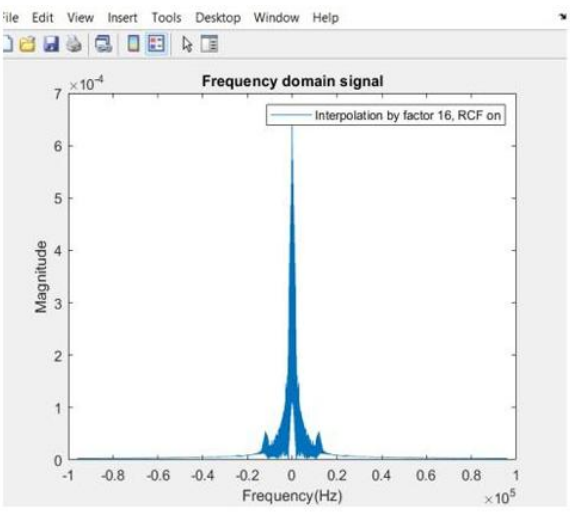
Applying interpolation by (16) without reconstruction filter:





Applying interpolation by (16) with reconstruction filter:





Interpolation Rate = 2

Time Domain

- **Without Filter:** The signal is twice as long, but the changes between points look sudden and unnatural.
- **With Filter:** The signal looks smooth, although the filtering process might cause a tiny delay in the wave.

Frequency Domain

- **With Filter:** The frequency graph looks almost exactly like the original.
- **Without Filter:** One small "ghost" signal (spectral image) appears near the original sampling frequency.

Interpolation Rate = 16

Time Domain

- **With Filter:** The signal is incredibly smooth and full of detail. It looks like a high-definition version of the original wave.
- **Without Filter:** The signal is mostly empty space. It is made of long flat lines or zeros with very few actual data points, making it look broken.

Frequency Domain

- **With Filter:** The spectrum stays perfectly clean and stays within the original frequency limits.
- **Without Filter:** Fifteen separate "ghost" images appear in the spectrum. Most of the signal's energy is now in the wrong places, completely ruining the data.